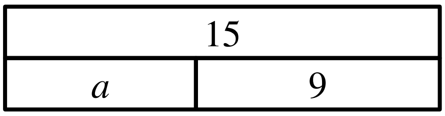
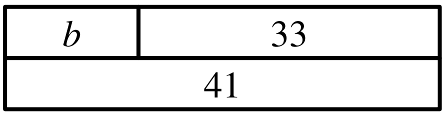
Solving simple linear equations with an additive step

Task A: Additive steps using representations

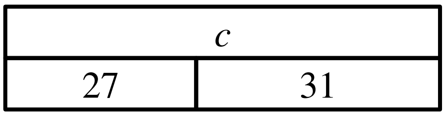
1) Find the value of the unknown in each bar model. Explain how you know.

a) 

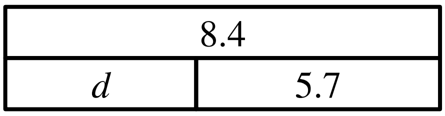
This represents the equation:
$15 - 9 = a$
$6 = a$ *solution when* a *is* 6

b) 

This represents the equation:
$b = 41 - 33$
$b = 8$ *solution when* b *is* 8

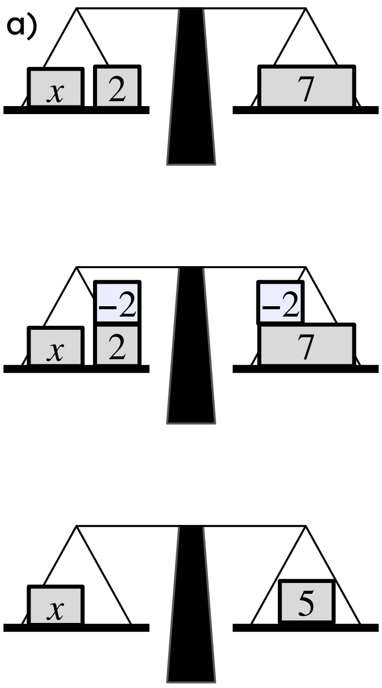
c) 

This represents the equation:
$c = 27 + 31$
$c = 58$ *solution when* c *is* 58

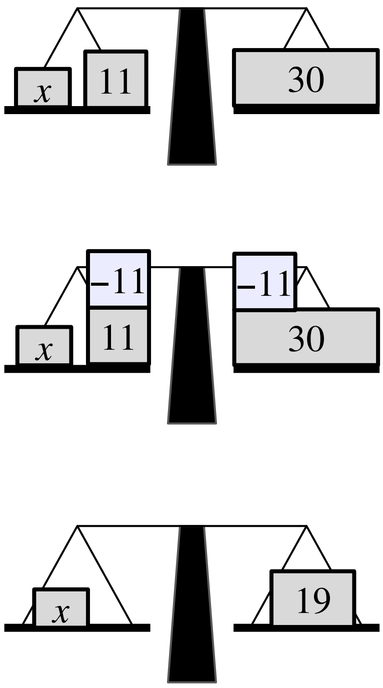
d) 

This represents the equation:
$8.4 - 5.7 = d$
$2.7 = d$ *solution when* d *is* 2.7

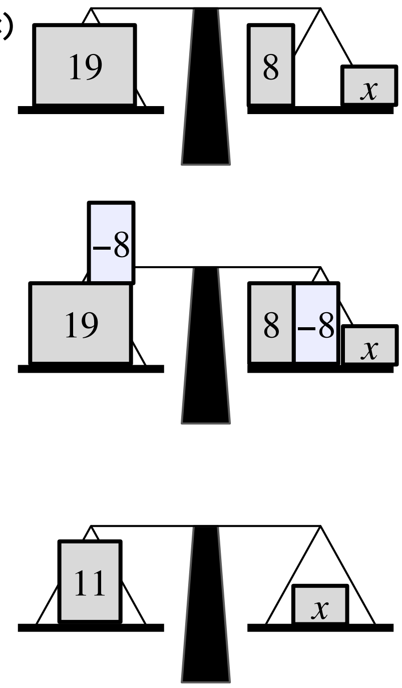
2) Complete the balance scales to show the additive step needed to find the solution.
What is the solution for each equation?

a) 

solution when x *is* 5

b) 

solution when x *is* 19

c) 

solution when x *is* 11

Task B: Writing solutions algebraically

1) Solve the following equations.

a)

$$\begin{aligned} x + 1 &= 12 \\ x + 1 + (-1) &= 12 + (-1) \\ x &= 11 \end{aligned}$$

d)

$$\begin{aligned} x + 13 &= 11 \\ x + 13 + (-13) &= 11 + (-13) \\ x &= -2 \end{aligned}$$

b)

$$\begin{aligned} x + 3.6 &= 17.8 \\ x + 3.6 + (-3.6) &= 17.8 + (-3.6) \\ x &= 14.2 \end{aligned}$$

e)

$$\begin{aligned} x - 10 &= -8.4 \\ x - 10 + 10 &= -8.4 + 10 \\ x &= 1.6 \end{aligned}$$

c)

$$\begin{aligned} x - 18 &= 42 \\ x - 18 + 18 &= 42 + 18 \\ x &= 60 \end{aligned}$$

f)

$$\begin{aligned} x + \frac{3}{5} &= -\frac{9}{10} \\ x + \frac{3}{5} + \left(-\frac{3}{5}\right) &= -\frac{9}{10} + \left(-\frac{3}{5}\right) \\ x &= -\frac{9}{10} - \frac{6}{10} \\ x &= -\frac{15}{10} \\ x &= -\frac{3}{2} \end{aligned}$$

2) Spot the mistakes in the following methods to solve these equations.

a)

$$\begin{aligned} x + 7 &= -3 \\ x + 7 + (7) &= -3 + (7) \\ x &= 4 \end{aligned}$$

Should have added −7 not 7 $x = -10$

c)

$$\begin{aligned} x - 5 &= 10 \\ x - 5 + 10 &= 10 + 10 \\ x + 5 &= 20 \end{aligned}$$

Hasn't isolated x, could now add −5 to both sides
$x = 15$

b)

$$\begin{aligned} x + 7.1 &= 23.5 \\ x + 7.1 + (-7.1) &= 23.5 \\ x &= 23.5 \end{aligned}$$

Forgot to add −7.1 to RHS $x = 16.4$

d)

$$\begin{aligned} x - 23 &= 23 \\ x - 23 + 23 &= 23 - 23 \\ x &= 0 \end{aligned}$$

Subtracted 23 instead of added 23 to RHS $x = 46$